

This report presents a four-year analysis (2014–2017) of 9,994 transactions from a US-based retail superstore, covering 5,009 orders across 793 unique customers and four geographic regions. The analysis was conducted as part of the Future Interns Data Science & Analytics Internship (Task 1) and is intended to inform strategic decisions around pricing, product mix, regional expansion, and discount policy.

Key Performance Indicators

\$2,297,201	\$286,397	12.5%	793
Total Revenue	Total Profit	Profit Margin	Unique Customers

Revenue & Profit Growth

Revenue grew from \$484,247 in 2014 to \$733,215 in 2017, a 51% increase over the period. Profit grew even faster — 89% over the same period — reflecting improving operational efficiency and a gradual shift toward higher-margin products. Q4 (October–December) is consistently the strongest quarter every year, accounting for approximately one-third of annual revenue, a pattern reliable enough to drive inventory and marketing planning.

Category Performance

Technology and Office Supplies are the profit engines of the business, each delivering margins above 17%. Furniture, despite comparable revenue to Technology, earns a margin of just 2.5%. Within Furniture, the Tables sub-category has accumulated a net loss of \$17,725 over the four-year period, and Bookcases a further \$3,473 — both driven by deep discounting rather than underlying demand weakness.

Category	Revenue	Profit	Margin
Technology	\$836,154	\$145,455	17.4%
Office Supplies	\$719,047	\$122,491	17.0%
Furniture	\$742,000	\$18,451	2.5%

Regional Performance

The West region leads in both absolute profit (\$108,418) and margin (14.9%) and warrants further investment. The Central region is the clear underperformer: despite \$501K in revenue, its 7.9% margin is nearly half that of the West, driven primarily by aggressive discounting on Binders in the Illinois–Texas corridor.

Region	Revenue	Profit	Margin
West	\$725,458	\$108,418	14.9%
East	\$678,781	\$91,523	13.5%
South	\$391,722	\$46,749	11.9%
Central	\$501,240	\$39,706	7.9%

Discount Policy

Discounting is the single largest controllable drag on profitability. 97% of all orders with a discount above 20% generated negative profit. Orders with no discount averaged \$42.10 profit per line item; orders with a discount above 40% averaged a loss of \$108.50. A hard cap at 20% — with exceptions requiring management approval — is the highest-impact, lowest-effort change available to the business.

Recommendations

- **Cap discounts at 20%.** Implement immediately as a hard policy ceiling. Any exception requires VP approval. Estimated annual profit recovery: \$50–\$80K.
- **Audit Tables and Bookcases.** Both are loss-making sub-categories. Evaluate a 10–15% price increase, supplier renegotiation, or SKU rationalisation.
- **Invest in Technology.** The highest-margin category (17.4%), led by Copiers (\$55,618 profit) and Phones (\$44,516). Prioritise inventory and marketing, especially ahead of Q4.
- **Expand in the West; fix Central.** Allocate additional sales headcount to the West (14.9% margin). Conduct a Central-region discount audit to identify and correct root causes.
- **Plan for Q4.** Use historical data to forecast demand by category and region, and commit inventory and budgets by September each year.